

FNCE 5352 — Lecture 4: Linear Regression (Intro + Interpretation)

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Tonight (3 hours)

- Linear regression as a *first* interpretable model
- Interpretation first, mechanics light
- Model assessment: *start the habits*

Where we are

- Data workflow + returns (Lecture 1)
- Tidy data, EDA, transformation (Lectures 2–3)
- Prediction vs inference, overfitting, train/test (Lecture 3)

Simple Linear Regression

The model (one predictor)

$$Y = \beta_0 + \beta_1 X + \varepsilon$$

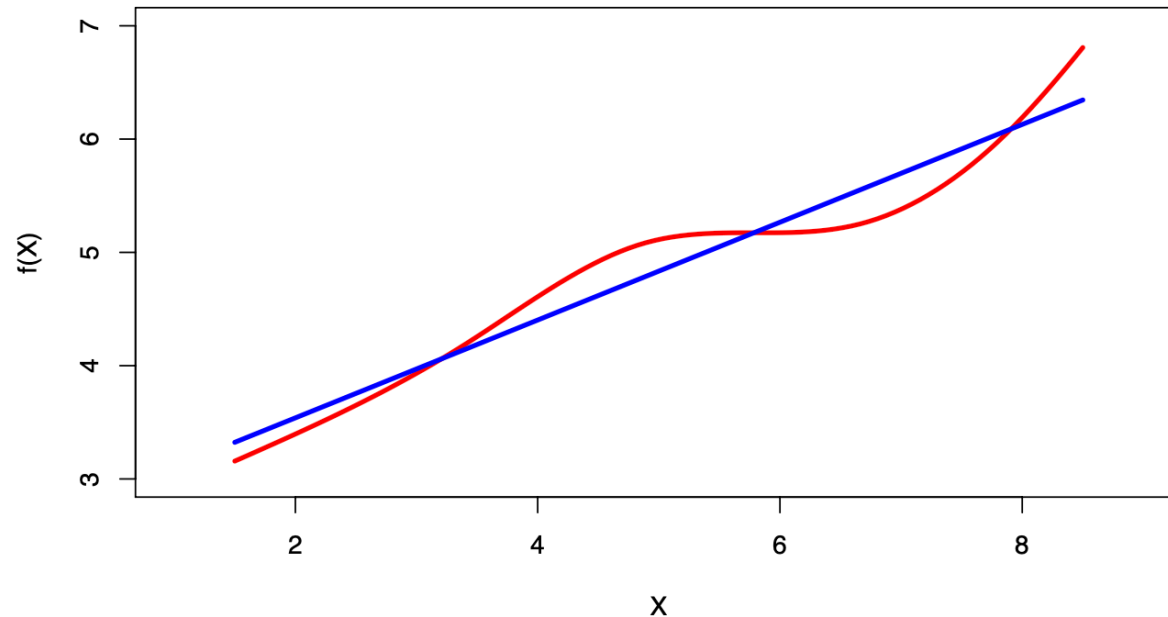
- slope = change in *expected* Y for +1 unit X
- intercept = expected Y when $X = 0$ (sometimes not meaningful)
- residual = what's left

Example 1: Advertising (clean lab)

- outcome: **Sales**
- predictor: **TV** (ad spend)
- question: "How much does Sales change when TV spend increases?"

Linear regression

- Linear regression is a simple approach to supervised learning. It assumes that the dependence of Y on $X_1, X_2, \dots, X_p$ is linear.
- True regression functions are never linear!



- although it may seem overly simplistic, linear regression is extremely useful both conceptually and practically.

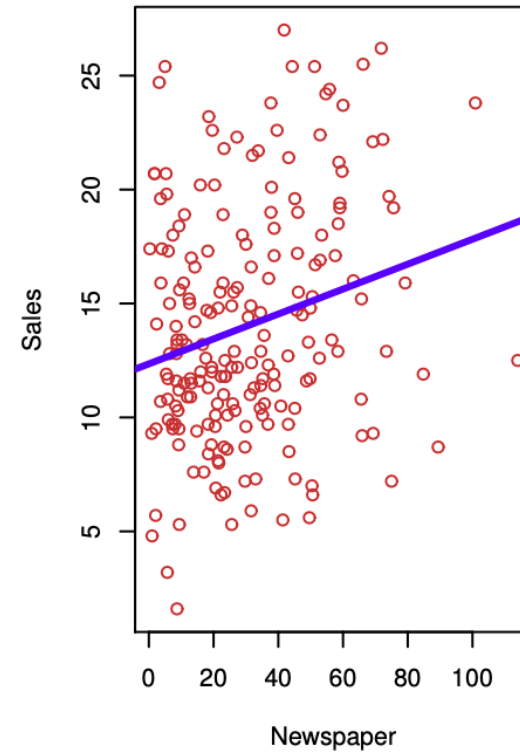
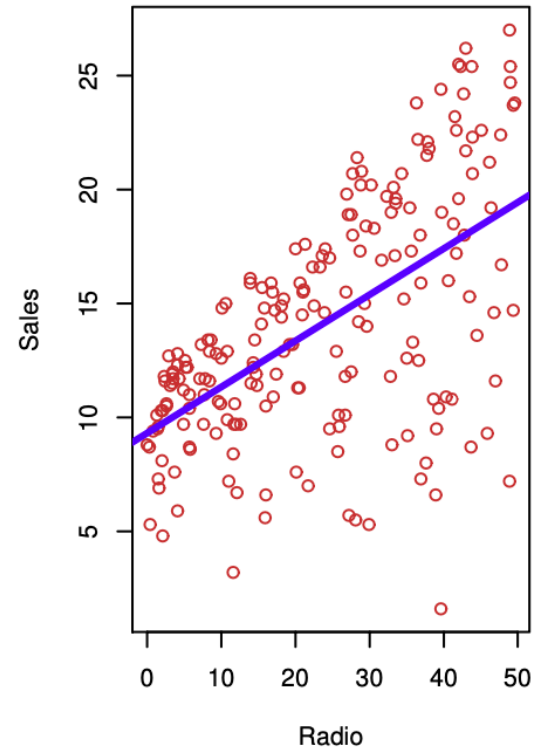
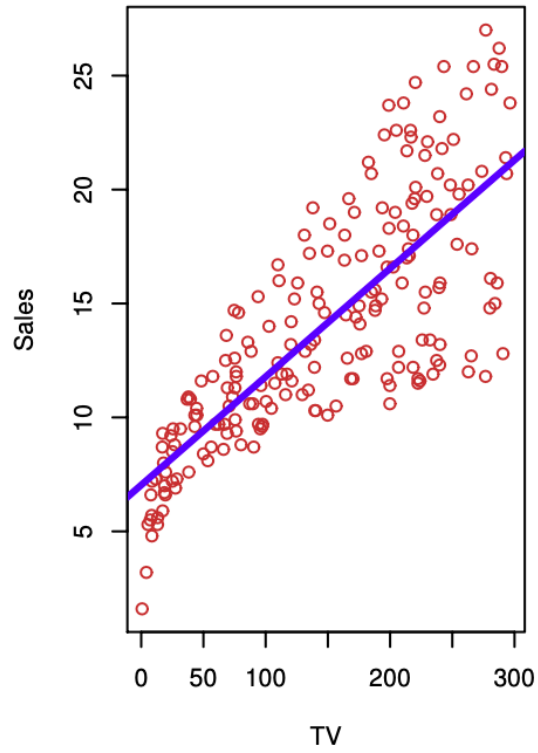
Linear regression for the advertising data

Consider the advertising data shown on the next slide.

Questions we might ask:

- Is there a relationship between advertising budget and sales?
- How strong is the relationship between advertising budget and sales?
- Which media contribute to sales?
- How accurately can we predict future sales?
- Is the relationship linear?
- Is there synergy among the advertising media?

Advertising data



Simple linear regression using a single predictor X .

- We assume a model

$$Y = \beta_0 + \beta_1 X + \epsilon,$$

where β_0 and β_1 are two unknown constants that represent the *intercept* and *slope*, also known as *coefficients* or *parameters*, and ϵ is the error term.

- Given some estimates $\hat{\beta}_0$ and $\hat{\beta}_1$ for the model coefficients, we predict future sales using

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x,$$

where $\hat{y}$ indicates a prediction of Y on the basis of $X = x$. The *hat* symbol denotes an estimated value.

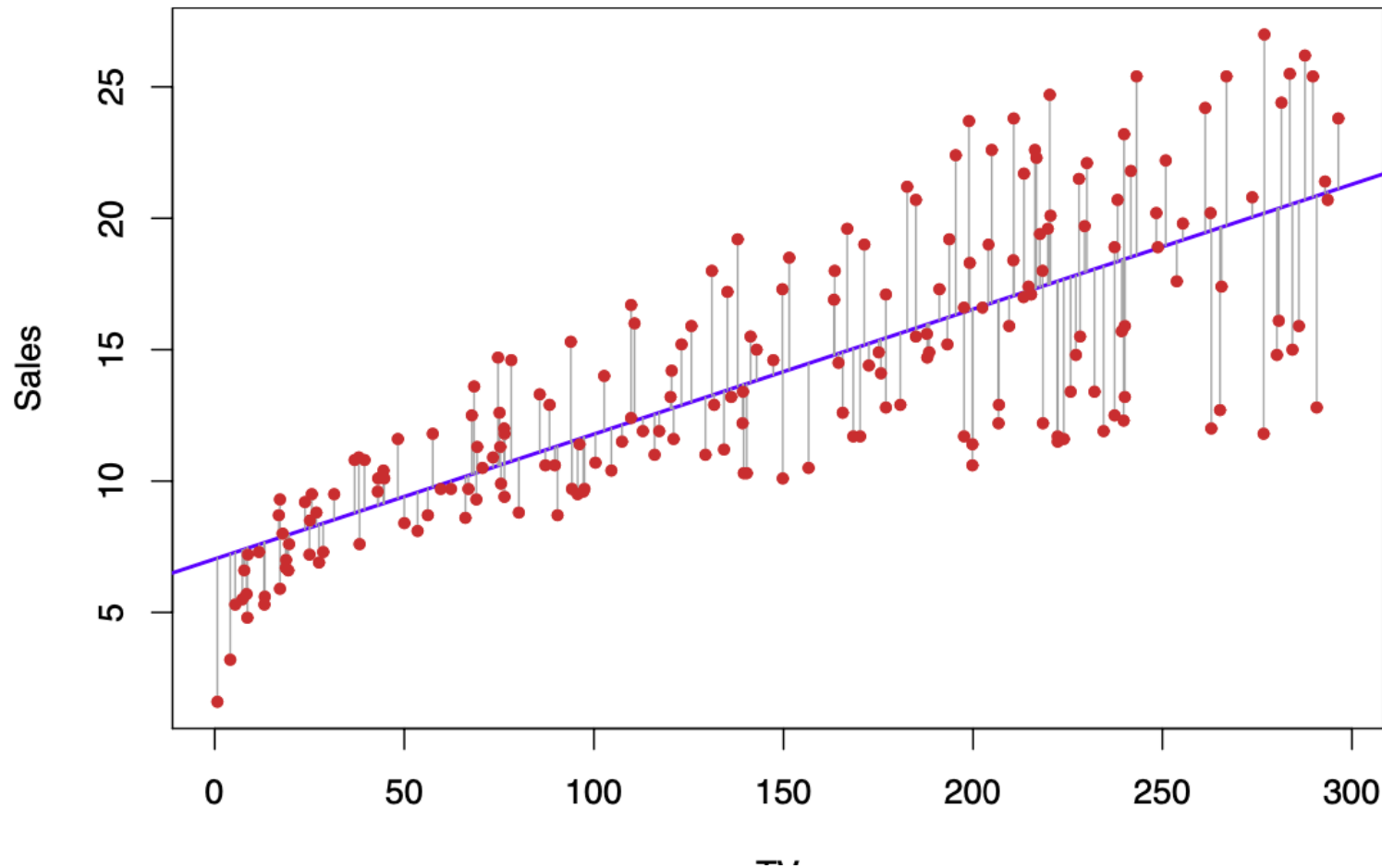
Example 2: Market Model

$$r_{AAPL,t} = \alpha + \beta r_{SPY,t} + \varepsilon_t$$

- β : "market sensitivity" (beta)
- α : "abnormal return" (in-sample; be humble)
- ε : idiosyncratic return (news / events / missing factors)

“Best line” idea

Regression chooses coefficients that make residuals “small” overall (by minimizing squared errors).



What $\hat{\beta}_0$ and $\hat{\beta}_1$ represent

- chosen from the data
- depend on the scale/units of X and Y
- change if you add/remove variables

Interpreting a slope (always with units)

Advertising: "+1 unit TV spend → expected Sales changes by $\hat{\beta}_1$ units"

Finance: "+1% SPY return → expected AAPL return changes by $\hat{\beta}_1$ % (same frequency)"

Quick check-in (1–2 minutes)

If $\beta = 1.20$, then:

“When SPY is up 1%, the model expects AAPL to be up about ____% (on average).”

Accuracy of coefficients

Coefficients come with uncertainty

We usually look at:

- standard errors (SE)
- confidence intervals (CI)
- t-stat / p-value (conceptual: "is slope plausibly 0?")

Results for the advertising data

	Coefficient	Std. Error	t-statistic	p-value
Intercept	7.0325	0.4578	15.36	< 0.0001
TV	0.0475	0.0027	17.67	< 0.0001

Accuracy of the model

Model accuracy (big picture)

Two different questions:

1. **In-sample fit:** did we explain anything?
2. **Generalization:** does it work on new data? (after break)

RSE, RMSE, and (R^2)

- **RMSE**: typical error size (units of Y)
- **RSE**: regression “typical error” (close cousin to RMSE)
- R^2 : fraction of variance explained (in-sample)

Assessing the Overall Accuracy of the Model

- We compute the *Residual Standard Error*

$$\text{RSE} = \sqrt{\frac{1}{n-2} \text{RSS}} = \sqrt{\frac{1}{n-2} \sum_{i=1}^n (y_i - \hat{y}_i)^2},$$

where the *residual sum-of-squares* is $\text{RSS} = \sum_{i=1}^n (y_i - \hat{y}_i)^2$.

- *R-squared* or fraction of variance explained is

$$R^2 = \frac{\text{TSS} - \text{RSS}}{\text{TSS}} = 1 - \frac{\text{RSS}}{\text{TSS}}$$

where $\text{TSS} = \sum_{i=1}^n (y_i - \bar{y})^2$ is the *total sum of squares*.

- It can be shown that in this simple linear regression setting that $R^2 = r^2$, where r is the correlation between X and Y :

$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}}.$$

Advertising data results

Quantity	Value
Residual Standard Error	3.26
R^2	0.612
F-statistic	312.1

Multiple linear regression

Multiple regression = same idea, more X's

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \varepsilon$$

Key phrase:

“Holding other variables constant”

Advertising: add more channels

$$Sales = \beta_0 + \beta_{TV} TV + \beta_{Radio} Radio + \beta_{News} Newspaper + \varepsilon$$

- Each slope is a *partial* relationship
- "TV effect, controlling for Radio and Newspaper"

Finance: add volatility context (VIX)

$$r_{AAPL,t} = \alpha + \beta_m r_{SPY,t} + \beta_v \Delta VIX_t + \varepsilon_t$$

- β_m : market sensitivity *controlling for VIX moves*
- β_v : return response to volatility changes *holding market fixed*

Interpreting coefficients (the catch)

- Works cleanly when predictors aren't strongly correlated
- When predictors move together, coefficient "stories" get slippery

Multiple Linear Regression

- Here our model is

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p + \epsilon,$$

- We interpret β_j as the *average* effect on Y of a one unit increase in X_j , *holding all other predictors fixed*. In the advertising example, the model becomes

$$\text{sales} = \beta_0 + \beta_1 \times \text{TV} + \beta_2 \times \text{radio} + \beta_3 \times \text{newspaper} + \epsilon.$$

Interpreting regression coefficients

- The ideal scenario is when the predictors are uncorrelated — a *balanced design*:
 - Each coefficient can be estimated and tested separately.
 - Interpretations such as “*a unit change in X_j is associated with a β_j change in Y , while all the other variables stay fixed*”, are possible.
- Correlations amongst predictors cause problems:
 - The variance of all coefficients tends to increase, sometimes dramatically
 - Interpretations become hazardous — when X_j changes, everything else changes.
- *Claims of causality* should be avoided for observational data.

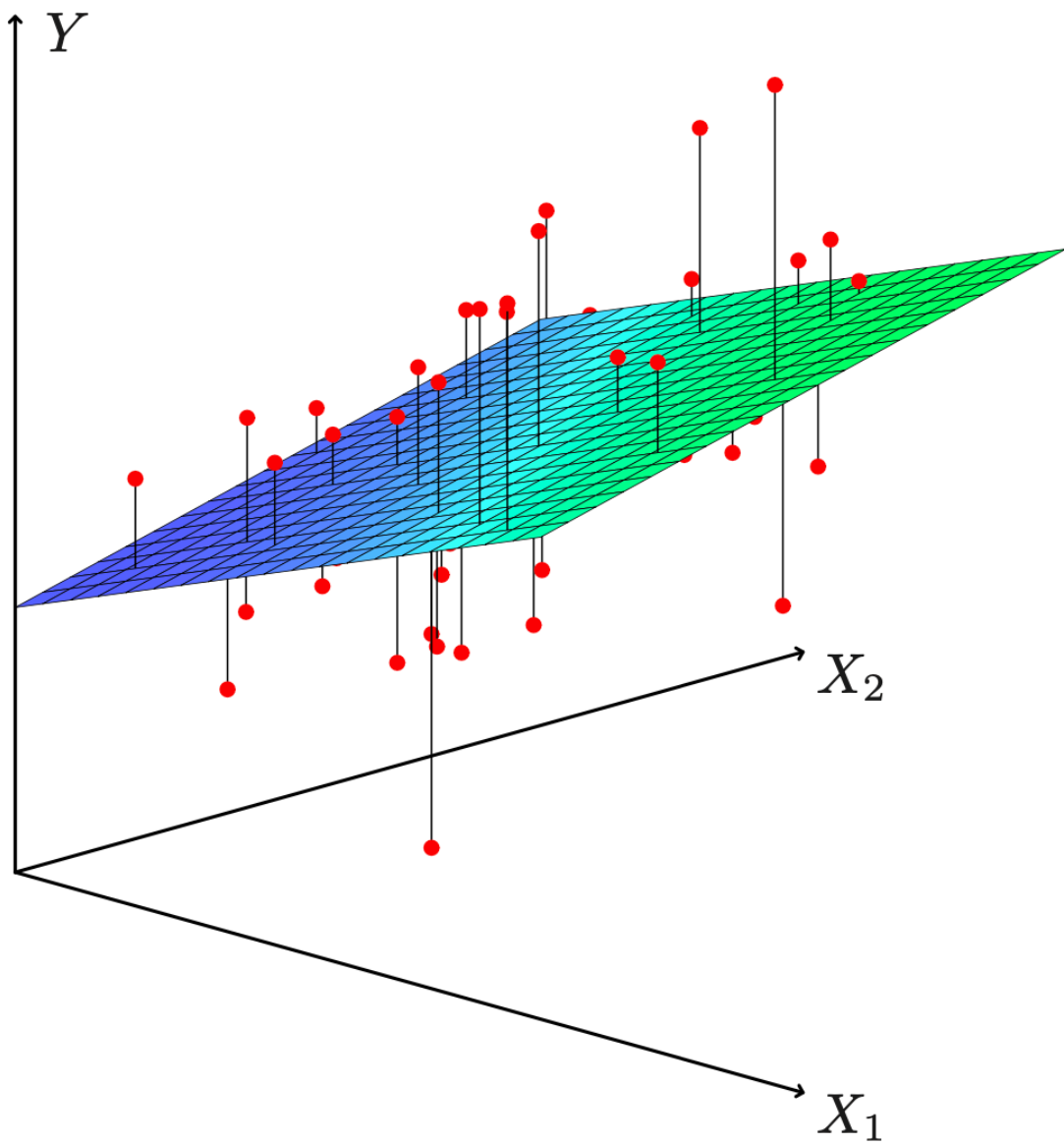
Two quotes by famous Statisticians

“Essentially, all models are wrong, but some are useful”

George Box

“The only way to find out what will happen when a complex system is disturbed is to disturb the system, not merely to observe it passively”

Fred Mosteller and John Tukey, paraphrasing George Box



Results for advertising data

	Coefficient	Std. Error	t-statistic	p-value
Intercept	2.939	0.3119	9.42	< 0.0001
TV	0.046	0.0014	32.81	< 0.0001
radio	0.189	0.0086	21.89	< 0.0001
newspaper	-0.001	0.0059	-0.18	0.8599

	Correlations:			
	TV	radio	newspaper	sales
TV	1.0000	0.0548	0.0567	0.7822
radio		1.0000	0.3541	0.5762
newspaper			1.0000	0.2283
sales				1.0000

Some important questions

1. *Is at least one of the predictors $X_1, X_2, \dots, X_p$ useful in predicting the response?*
2. *Do all the predictors help to explain Y , or is only a subset of the predictors useful?*
3. *How well does the model fit the data?*
4. *Given a set of predictor values, what response value should we predict, and how accurate is our prediction?*

Is at least one predictor useful?

For the first question, we can use the F-statistic

$$F = \frac{(\text{TSS} - \text{RSS})/p}{\text{RSS}/(n - p - 1)} \sim F_{p, n-p-1}$$

Quantity	Value
Residual Standard Error	1.69
R^2	0.897
F-statistic	570

What we'll do interactively (after break)

- Advertising: simple $\rightarrow$ multiple regression
- AAPL/SPY: market model beta estimate
- Add VIX: multiple regression interpretation
- Metrics: RMSE/MSE/ R^2 (including `yardstick`)

Break

15 minutes